1 **(a)** Correct triangle B with vertices at $(-3,3),(-1,3),(-1,6)$ **[B2]**

Translation by vector $\begin{pmatrix}-4\\k\end{pmatrix}$ or $\begin{pmatrix}k\\2\end{pmatrix}$ **[B1]**

(b) Correct triangle C with vertices at $(-3,1),(-5,1),(-5,4)$ **[B2]**

Reflection in $y=-1$ or correct orientation but wrong position **[B1]**

[Total Marks: 4]

2 **(a)** Correct translation of Shape A **[B2]**

If incorrect, translation of $\begin{pmatrix}-4\\k\end{pmatrix}$ or $\begin{pmatrix}k\\2\end{pmatrix}$ where $k\neq 0$ **[B1]**

(b) Correct reflection of Shape A to form Shape C with vertices at $(1,-3)$, $(3,-3)$, and $(1,-6)$ **[B2]**

If incorrect, reflection in $x=-1$ or $y=1$ **[B1]**

[Total Marks: 4]

3 **(a)** Correct triangle with vertices at $(-3,3),(-1,3)$ and $(-1,6)$ **[B2]**

Translation by correct vector but incorrect shape/orientation OR correct size, shape and orientation but wrong translation **[B1]**

(b) Reflection **[B1]**

in the y-axis OR $x=0$ **[B1]**

[Total Marks: 4]

4 **(a)** Vertices of image plotted correctly at $(-1,-4)$, $(1,-4)$ and $(1,-1)$ **[B2]**

If not B2, award B1 for correct size and orientation, or 2 vertices correct **[B1]**

(b) Reflection **[B1]**

in the y-axis (or $x=0$) **[B1]**

[Total Marks: 4]

5 **(a)** Image B drawn correctly at $(-4,3),(-2,3),(-2,6)$ **[B2]**

If incorrect, 1 mark for translation by $\begin{pmatrix}-5\\k\end{pmatrix}$ or $\begin{pmatrix}k\\2\end{pmatrix}$ **[M1]**

(b) Image C drawn correctly at $(1,-1),(3,-1),(3,-4)$ **[B2]**

If incorrect, 1 mark for reflection in y-axis or in a line $y=k$ **[M1]**

(c) Image D drawn correctly at $(-1,-1),(-3,-1),(-3,-4)$ **[B2]**

If incorrect, 1 mark for rotation of $180°$ about the wrong centre, or $90°$ about $(0,0)$ **[M1]**

[Total Marks: 6]

6 **(a)** Correct shape with vertices at $(-2,-4),(-6,-4),(-6,-6),(-4,-6),(-4,-8)$ and $(-2,-8)$ **[B2]**

Enlargement, scale factor 2 centre $(0,0)$ OR correct scale factor and orientation but wrong centre **[B1]**

(b) Correct shape with vertices at $(4,-1),(4,-3),(5,-3),(5,-2),(6,-2)$ and $(6,-1)$ **[B2]**

Rotation of 90° clockwise about the wrong centre OR rotation of 90° anti-clockwise about $(1,-1)$ **[B1]**

[Total Marks: 4]

7 **(a)** Rotation **[B1]**

90° clockwise OR 270° anti-clockwise **[B1]**

Centre $(0,0)$ **[B1]**

(b) Correct triangle drawn with vertices at $(0,2)$, $(1.5,2)$, and $(1.5,3)$ **[B2]**

If incorrect, correct size and orientation but wrong position OR correct centre of enlargement but scale factor 2 **[B1]**

[Total Marks: 5]

8 **(a)** Image B drawn correctly at $(2,-4),(2,-2),(6,-2)$ **[B2]**

If incorrect, 1 mark for reflection in $y=-x$ or correct orientation but wrong position **[M1]**

(b) Image C drawn correctly at $(-4,2),(-2,2),(-2,-2)$ **[B2]**

If incorrect, 1 mark for rotation 90° anti-clockwise about $(2,2)$ or 90° clockwise about the wrong centre **[M1]**

(c) Reflection **[B1]**

in the line $y=2$ **[B1]**

[Total Marks: 6]

9 **(a)** Vertices of image plotted correctly at $(-1,2)$, $(-2,2)$ and $(-2,0.5)$ **[B2]**

If not B2, award B1 for enlargement SF $-\frac{1}{2}$ from wrong centre, or correct centre but SF $\frac{1}{2}$ **[B1]**

(b) Rotation **[B1]**

180° (direction not required) **[B1]**

centre $(1,2)$ **[B1]**

(c) Calculates area of $T = \frac{1}{2} \times 2 \times 3 = 3$ **[M1]**

Sets up $3 \times k^2 = 48 \implies k^2 = 16$ **[M1]**

$k = 4$ **[A1]**

[Total Marks: 7]

10 **(a)** Enlargement **[B1]**

Scale factor -2 **[B1]**

Centre $(0, 1)$ **[B1]**

(b) Enlargement scale factor $-\frac{1}{2}$ **[B1]**

Centre $(0, 1)$ **[B1]**

(c) Translating Shape T gives at least two correct coordinates of V: $(-2, 3)$, $(-1, 3)$, $(-1, 5)$ **[M1]**

Recognising reflection in $y = x$ swaps coordinates (x, y)

(y, x)$[M1]$

Final coordinates: $(3, -2)$, $(3, -1)$, $(5, -1)$ **[A1]**

[Total Marks: 8]

11 **(a)** Rotation **[B1]**

$90°$ anticlockwise OR $270°$ clockwise **[B1]**

Centre $(0, 0)$ **[B1]**

(b) Correct shape R with vertices at $(4, 4), (5, 4), (5, 5), (4.5, 5), (4.5, 4.5), (4, 4.5)$ **[B2]**

Any enlargement scale factor $\frac{1}{2}$ OR correct centre $(6, 6)$ with incorrect scale factor **[B1]**

[Total Marks: 5]

12 **(a)** Vertices of image plotted correctly at $(1, 1)$, $(2, 1)$, $(2, 1.5)$, $(1.5, 1.5)$, $(1.5, 2)$ and $(1, 2)$ **[B2]**

If not B2, award B1 for correct size and orientation, or correct enlargement from the wrong centre **[B1]**

(b) Rotation **[B1]**

$90°$ clockwise (or $-90°$) **[B1]**

centre $(0, 0)$ (or origin) **[B1]**

[Total Marks: 5]

13 **(a)** Enlargement **[B1]**

Centre $(0, 0)$ or origin **[B1]**

Scale factor $-\frac{1}{2}$ **[B1]**

(b) Image R drawn correctly at $(1, 3), (3, 3), (3, 4)$ **[B3]**

If incorrect, 2 marks for correct size and orientation but wrong position **[M2]**

1 mark for correct centre used but wrong scale factor **[M1]**

(c) Rotation **[B1]**

$180°$ **[B1]**

Centre $(0, 1)$ **[B1]**

OR Enlargement [B1], Scale factor -1 [B1], Centre $(0, 1)$ **[B1]**

[Total Marks: 9]

14 **(a)** Enlargement **[B1]**

Scale factor $-\frac{1}{2}$ **[B1]**

Centre $(0, 0)$ **[B1]**

(b) Reflection **[B1]**

in the x-axis OR $y = 0$ **[B1]**

[Total Marks: 5]

15 **(a)** Correct triangle U with vertices at $(1, 2), (3, 2)$ and $(3, 0)$ **[B2]**

Enlargement scale factor $\frac{1}{2}$ from wrong centre OR ray lines drawn from $(0, 4)$ but scale factor incorrect **[B1]**

(b) Correct triangle V with vertices at $(2, 1), (2, 3)$ and $(0, 3)$ **[B2]**

Reflection of U in the line $y = -x$ OR reflection of T in the line $y = x$ **[B1]**

(c) Enlargement **[B1]**

Scale factor 2 **[B1]**

Centre $(0, 4)$ **[B1]**

[Total Marks: 7]